

Higher Levels of IGF-I and Adrenal Androgens at Age 8 Years Are Associated with Earlier Age at Menarche in Girls

Context: Earlier age at menarche is associated with rapid infancy weight gain and childhood obesity. The role of hormone levels in mediating these associations is unclear. Objective: The aim of this study was to identify childhood hormone levels at age 8 yr that are associated with early menarche, independent of body size. Design, Settings, and Subjects: A total of 329 girls from a prospective United Kingdom birth cohort study provided blood samples at mean age 8.1 yr (range, 8.0–8.5) for hormone measurements and were followed longitudinally to establish age at menarche. Main Outcome Measures: Fasting plasma levels of IGF-I, androstenedione, dehydroepiandrosterone sulfate (DHEAS), leptin, insulin, IGF binding protein-1, and SHBG were measured. Age at menarche was reported by questionnaire and categorized as before 12.0, 12.0–13.0, or later than 13 yr. Results: Earlier menarche was associated with greater body weight, height, and body mass index at age 8 yr (all P-trend <0.001). Before adjustment for body size, earlier menarche was associated with higher levels of IGF-I, androstenedione, DHEAS, leptin, and fasting insulin, and with lower levels of IGF binding protein-1 and SHBG at age 8 yr (all $P < 0.01$). After adjustment for body mass index and height at age 8 yr, only IGF-I ($P = 0.004$), androstenedione ($P = 0.01$), and DHEAS ($P = 0.01$) remained associated with earlier menarche. Conclusions: Associations between higher levels of IGF-I and adrenal androgens at age 8 yr with earlier menarche, independent of body size, support functional roles of these hormones in regulating puberty timing in girls. Higher levels of these hormones reported in children who exhibited rapid weight gain during infancy may indicate their role in developmental pathways leading to earlier sexual maturation.